

Exp No:16 SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

Github: <https://github.com/Pardhav2006/embedded-system>

Program:

```
#include <reg51.h>

#include <stdio.h>

void serial_ISR(void) interrupt 4
{
    char ch;

    if (RI)
    {
        RI = 0;    // Clear receive interrupt flag

        ch = SBUF;

        // Read received character

        printf("\n>>> INTERRUPT RECEIVED: ");

        printf("%c\n", ch);
    }
}

void delay(void)
{
    unsigned int i;

    for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled

    TMOD = 0x20; // Timer1 mode 2 (auto-reload)

    TH1 = 0xFD; // Baud rate 9600

    TR1 = 1;    // Start Timer1
```

```
IE = 0x90;    // EA = 1, ES = 1 (enable serial interrupt)
```

```
TI = 1;
```

```
while(1)
```

```
{
```

```
// Enable printf
```

```
printf("Hello World\n");
```

```
delay();
```

```
}
```

```
}
```

